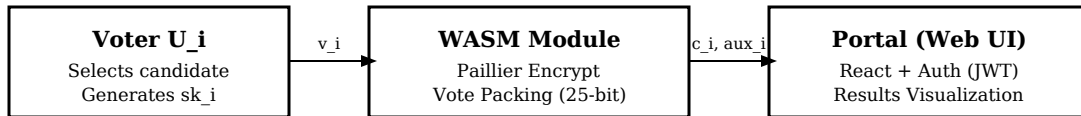


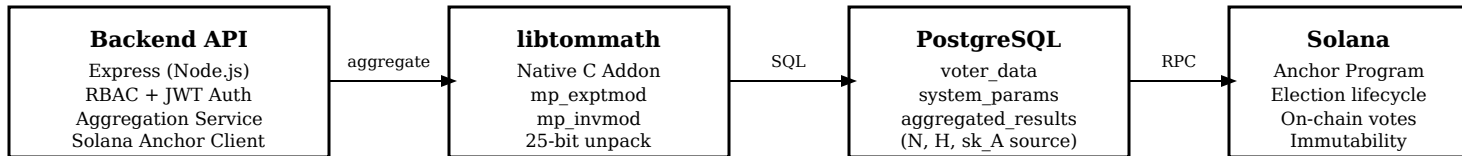
Proposed System Architecture

Client Side (Browser)



HTTPS

Server Side (Containerized Microservices)



Cryptographic Parameter Flow:

system_params table → (N, H, sk_A) → Aggregation Service (single source of truth, NOT blockchain)
Solana → election metadata (candidates, stage, on-chain vote hash) → Backend → Portal

Trust Boundaries:

- TP is trusted only for setup (generates $N=pq$, knows factorization, then goes offline)
- Aggregator A knows sk_A but cannot learn individual votes (only the sum, under DCR)
- Collector C sees aux_i values but cannot decrypt (random under DCR)
- Voter U_i knows only own sk_i and vote — plaintext never leaves browser
- Solana validators are untrusted (Byzantine fault tolerant consensus)